GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Specification for Mainline Valves**
Document Number	EA-BFPCL-GPMC-GGPL-FEED- PPL-SPC-003
Document Revision	R01
Document Status	Issued for Review
Originator	E. Uwaoma
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Issue Date	8-Jun-22

Specification for Mainline Valves

R01	08.06.22	Issued for Review	E. Omeke	E. Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	08.06.22	First Issue for Client's Review
R02		
A01		

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
   for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED- PPL-SPC-003

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Wall Thickness Calculation Report
EA-BFPCL-GPMC-GGPL-FEED-PPL-ALS-001 to 036	Pipeline Alignment Sheet
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-010	Pipeline Cleaning, Gauging, & Hydrotesting Specification.
EA-BFPCL-GPMC-GGPL-FEED-PPL-PHY-001	Pipeline Integrity Management Philosophy.
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-009.	Specification for Pipeline Construction.

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Abbreviations

ABBREVIATIONS	MEANING
AED	Anti-Explosive Decompression
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
API	American Petroleum Institute
BS	British Standard
CS	Carbon Steel
DEP	Design and Engineering Practices
DP	Design Pressure
DWT	Drop Weight tear
ENP	Electroless Nickel Plated
FEED	Front End Engineering
ISO	International Organisation for Standardization
3LPE	Three-Layer Polyethylene
MESC	Material & Equipment Standard and Code
MMscf/d	Million standard cubic feet per day
MSS	Manufacturers Standardization Society
MTO	Material Take Off
NDE	Non-Destructive Examination
NPS	Nominal Pipe Size
NPT	Nominal Pipe Threaded
NPS	Nominal Pipe Size
OD	Outer Diameter
PE	Polyethylene
PSL	Product Specification Level
RF	Raised Face
RTJ	Ring Type Joint
SMTS	Specified Minimum Tensile Strength
SMYS	Specified Minimum Yield Strength
SPE	Society of Petroleum Engineers
SS	Stainless Steel
SW	Socket Weld
TD	Design Temperature
WT	Wall thickness
WPS	Welding Procedure Specification
WPQR	Welding Procedure Qualification Record

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

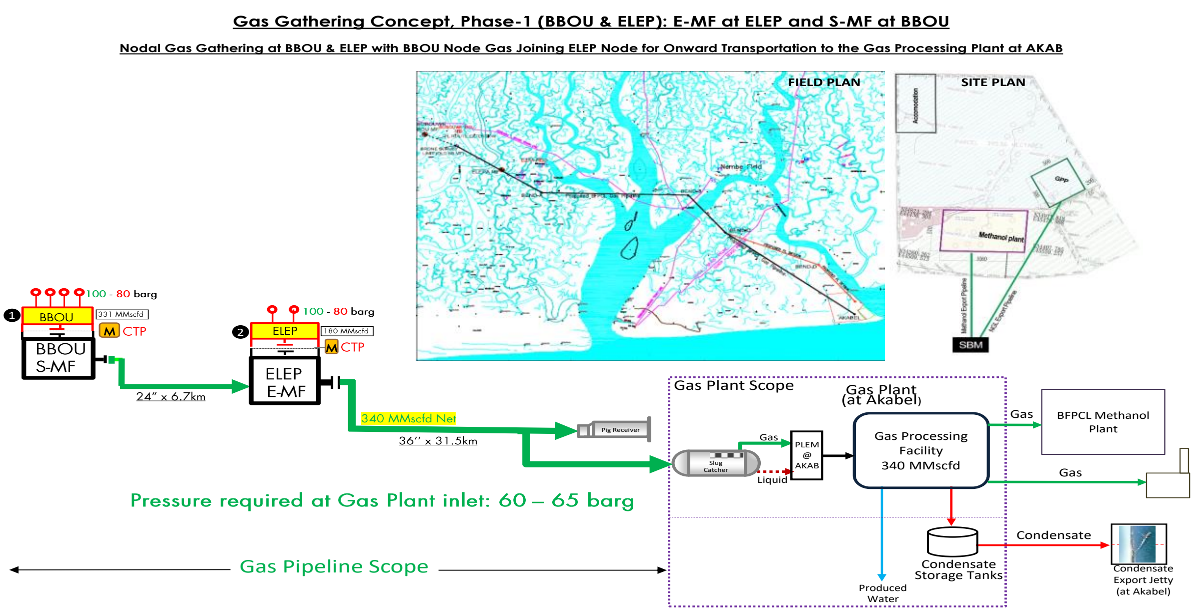

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification details the minimum requirements for manufacturing, inspection, testing, marking and delivery of and supply of Main line carbon steel Valves for phase 1 of BFPCL Gas Gathering Pipelines & Associated Infrastructure.

The FEED shall fully comply with all specified relevant contractual requirements The FEED

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 31.36.00.30-Gen.	Pipeline Transportation Systems - Pipeline Valves (Amendments/Supplements To API Spec 6D)
DEP. 82.00.10.10-Gen	Project Quality Assurance
MESC SPE 77/302	Valves – General requirements
SPE 77/130	Ball Valve to ISO 14313 / API 6D, Flanged or Butt Weld Ends
ASME B31.8	Gas Transmission and Distribution Piping Systems
ASME B16.5	Pipe Flanges and Flanged Fittings NPS ½ Through NPS 24
ASME B16.10	Face-to-Face and End-to-End Dimensions of Valves
ASME B16.34	Valves – Flanged, Threaded, and Welding End
ASME B31.8	Gas Transmission and Distribution Piping Systems
ASTM A703	Standard Specification for Steel Castings, General Requirements, for Pressure-Containing Parts
ASTM B 633	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
API Spec 6D	Specification for Pipeline Valves
API Spec 6 FA	Specification for Fire Test for Valves
API Spec 6B	Specification for Fire Test for End Connections
API 598	Valve Inspection and Testing
API 607	Fire Test for Quarter-Turn Valves and Valves Equipped with Non-metallic Seats
BS EN 10204	Metallic Products - Types of Inspection Documents
ISO 9001	Quality Systems Model for Quality Assurance in Design, Development, Production, Installation and Servicing

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Line Pipe Data

3.1. 24" Multiphase Pipeline Data

The 24" pipeline design data for the Multiphase Pipeline Project are as shown below

Table 3.1: 24" Multiphase Pipeline Data

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	**6.7km**
Design Life	30 years
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L–X70(PLS2) SAWL - 100% radiographed.
OD / Wall Thickness	**610/15.88mm**
Corrosion Allowance	3 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Installation Condition	Trenching
Cathodic Protection	Impressed current

3.2. 36" Multiphase Pipeline Design Data

The 36" pipeline design data for the Multiphase Pipeline Project are as shown below

Table 3.2: 36" Multiphase Pipeline Data

PARAMETER	DESCRIPTION
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	**31.5km**
Design Life	30 years
Process Fluid	Condensate/hydrated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature	-13.3°C
Line Pipe Material Grade	API 5L–X70 (PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	**36"**
OD / Wall Thickness	**914/23.83mm**
Corrosion Allowance	6 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Installation Condition	Trenching
Cathodic Protection	Impressed current

4. Environmental and Pipeline Design Data

Refer to the document EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002 - Basis of Design

5. Valve Requirements

5.1. General

Valves shall be designed and engineered for a service life of 30 years.

The design and construction shall comply with API 6D for pipeline ball and check valves and supplemented by DEP 31.36.00.30-Gen and Shell MESC Group 77. Additional project specific requirements shall be listed on the valve data sheets and the material requisitions.

Where Tight shut–off is specified, valve shall comply with seat leakage rates A according to ISO 5208

All valves shall be suitable for vacuum drying and vacuum service at pressures of 500 Pa (0.07 psia).

Lifting lugs shall be provided for valves with mass of 250kg and above.

Valve supports (legs or saddles) shall be supplied and installed on valves of nominal size 12 inches and above. The supports must allow access to body drain plugs where fitted.

All valves (whether operated with a lever or a gear operator) shall be provided with locking device plate to facilitate padlocks or car-seals.

All valves shall have position indicator and the design shall be such that the component(s) of the indicator and the valve operator cannot be assembled to falsely indicate the valve position.

Valve stem design shall be of blowout-proof type. Note that ball valve stem retention shall not depend on the packing gland or by means of body/stem threads.

Valves used in fluids containing solid particles shall have metal-to-metal seating.

All external connections and auxiliary fittings provided on the body of valves for various purposes shall be able to withstand the hydrostatic shell test pressure of the valve.

Bolting shall be provided with zinc plated / bichromate surface treatment. Cadmium plating is not allowed.

In addition to name plate, valve shall be supplied with a stamped SS316 tag fixed to the gland bolting or hand-wheel with SS316 wire indicating project information such as PO No., tag number, valve type, size, rating, painting system, item number reference to valve MTO attached to RFQ, etc.

Face to Face and End to End dimensions for full and reduced bore sizes of all valves shall comply with API Spec 6D / ISO 14313. Face-to-face and bore dimensions not covered by API Spec 6D shall be subject to prior agreement and be stated in the quotation.

5.2. Requirements and Limitation for Materials

Valve material selection shall be as specified in the valve datasheets and the projects piping material specification and classification.

No specified material substitution with an equivalent or higher material shall be allowed without a written authorization from EMPLOYER

Cast material cannot be proposed as substitute to forged material unless specifically agreed by the EMPLOYER. Unless otherwise specified steel casting for pressure containing parts shall comply with general requirements of ASTM A703 with all its supplementary requirements.

Cast material is not recommended for valve stem whatever the pressure class and valve size.

The selection of seal materials for ball valves shall be based on actual maximum and minimum operating temperatures of the valves, pressure, and fluid composition. Manufacturer shall ensure seals compliance to anti explosive decompression properties.

When specified that valves shall have weld overlay cladding, the welding shall be performed in accordance with MESC SPE 77/313. The additional welding and hard facing requirements contained in Clause 3 "Welding and hard facing requirements" of MESC SPE 77/302 shall be met

Seat insert shall be of the re-enforced polymeric material type.

Graphite seals are not acceptable as primary pressure containment seal. They shall only be used as back up seal for fire resistance properties.

The chemical composition and carbon content restriction for valve pressure containing parts (body, bonnet, cover) and stem shall be as detailed in MESC SPE 77/302.

5.3. Ball Valve Configuration

Ball valve shall be fully dismantled bolted design, top or side entry as indicated in datasheet.

For welded end valve, only top entry design shall be used.

Short pattern ball valves shall not be furnished

Valves obturator shall be one-piece forging and mounted on a trunnion, spherical in shape and rotates on an axis perpendicular to flow direction.

5.4. Check Valve Configuration

To achieve fugitive emission class A (HS), only dual plate check valves of retainerless design shall be used. Dual check valves if specified shall be designed as per API 6D.

Wafer-lug type dual plate check valves will be of the type where the bolting is continuously covered across the valve body by the lug, thereby protecting the bolting from direct fire impingement (in the event of a fire) or degradation by general corrosion.

When retainer less dual plate check valves are assembled and compressed by the mating flange, the seat retainer face shall not protrude beyond the body face.

The design for swing check shall also be retain less where the clapper and hinge pin are internally assembled. Swing check valve shall be as per API 6D

The body to bonnet connection for swing check shall be flanged and the flange facings shall be male-and-female, tongue-and-groove, or ring joint type. Flange sealing faces shall have a circular shape and shall have a facing finish in accordance with ASME B16.5

Swing check valves shall be piggable by having a have full opening bore

6. Design and Special Requirements

6.1. Valve Bore

Valves specified as full bore shall have internal opening not less that the size specified in table 1 of API 6D. Internal opening for reduced bore valve may be less what is specified in table 1. Note that there is no restriction to upper limit of valve bore size.

6.2. Design and calculations

The pressure/temperature rating of the valve shall be fully in accordance with the appropriate class for the body material in ASME B16.5 or ASME B16.34 for temperatures up to 150 °C (302 °F)

At higher temperature rating, the manufacturer shall consider the effect on selected sealing materials

All bolting calculations must satisfy ASME B16.34 requirements.

The valve Manufacturer shall submit a fully detailed calculation note that demonstrates that body and internal component stresses are within acceptable levels. Calculations must be based on the applicable code and standards

6.3. Valve End

6.3.1. Flanged Ends

Standard end flanges shall be furnished with a raised face as specified in the datasheet and designed in accordance with ASME B16.5 for diameters up to 24”. The flange facings shall undergo finish machining in accordance with MSS SP-6

The minimum Brinell hardness of grooves of RF flanges shall be 110 for Carbon Steel flanges

6.3.2. Welding Ends

Welding end preparation if specified, shall conform to figure I-4 and I-5 of ASME B31.8. The valve body shall be furnished with extension pieces (pup pieces) which

shall be considered as an integral part of the valve length. Length of pup piece shall be;

- for valves ≤ DN 300 (≤ NPS 12): 400 mm (16 in)
- for valves > DN 300 (> NPS 12): 800 mm (32 in)

The outside diameter, wall thickness, material grade and composition shall conform to the specification of the adjoining piping. Otherwise, wall thickness transition tapers shall not be steeper than 1:4 and comply with ASME B31.3, Figure 328.4.3. (ASME B31.8, Figure I-5, ISO 13847 Clause 7.7).

6.4. Seat Design

Seats rings shall be of the piston effect design type.

Seat shall be energized with springs to enable adequate sealing at low differential pressure and able to be line pressure energized at all differential pressures. The entrapment of solids will not obstruct free movement of the seat rings.

On split body and top entry valves, the seat design shall be such as to allow for easy trim removal (renewable)

6.5. Valve Cavity Pressure Relief

For double seated valves where liquids can be trapped in the body cavity while subjected to temperature changes and consequent expansion of the liquid, cavity relief functionality shall be provided as follows:

- Ball valves having single-piston effect seating shall internally relieve the excessive cavity pressure to the downstream (low pressure) side of the valve.
- Ball valves having double-piston effect shall be fitted with an external cavity relief valve.
- Ball valves having both a single and a double piston effect seating in the same valve, shall internally relieve the excessive cavity pressure via the single-piston effect seat.

Pressure relief shall not be achieved by using a pressure-equalising hole in the

obturator connecting the body cavity to the High Pressure (HP) side of the closure member.

6.6. Non-Metallic seals

The selection of seal materials shall be based on actual maximum and minimum operating temperatures of the valves, pressure, fluid composition and chemical product used on a temporary basis.

Seals shall provide tightness to achieve a minimum of fugitive emission class B as specified in MESC SPE 77/312. 3. Fugitive emission production testing shall be conducted according to MESC SPE 77/312.

Generally, elastomeric seals shall comply with MESC SPE 85/301

6.7. Injection Points

Valve design and selection of materials should negate the need for the provision of sealant and lubricant injection connections. If specified, sealant injection for upper stem and seat seals shall be a drilled and threaded connection incorporating a non-return valve integral with the plug head.

6.8. Pigging Consideration

Full bore valves shall be capable of being pigged, sphered and scraped regularly without damage to the soft seat, if fitted. The bore of the valve in the open position, shall be verified for smooth passage of a pig or scraper for the entire length and diameter of the valve.

6.9. Fire Safe Requirement

When fire safe design is specified, graphitic or metallic seals shall be provided to protect the valve against leakage in case of a fire and by this feature the valve is capable of passing a fire test.

All valves shall be tested and certified fire safe in accordance with the procedures of API SPEC 6FA, ISO 10497.

6.10. Anti-Static Device

Valves shall incorporate suitable devices in their design to ensure electrical continuity between the obturator, stem, and body of the valve to avoid accumulation of stray current.

6.11. Drain outlet

All ball valves full-bore/reduced bore shall be provided with a drilled and threaded drain connection according to the table below.

Table 4.0—Thread/Pipe Sizes for Drains

Pressure class	Pipe Size(NPS)	Minimum Pipe Thread / Pipe Size(mm)
≤ ASME Class 900	2 to 8 in	15
	> 8 in	25

The profile for the thread shall be in accordance with ISO 228-1 and fitted with a threaded plug with primary seal inboard of the thread, in order to protect it against crevice corrosion.

The plug shall have an external shoulder suitably profiled to accept site seal weld 1/3 times the nominal size of the plug

Manufacturer shall ensure that there is material compatibility between valve body/trim and plug/ seal materials.

6.12. Operation of the valves

6.12.1. Manual operators

Generally, valves shall be operated by hand-wheel or by wrench. Hand-wheel design and dimensional limits shall be as per API 6D.

The sizing of valve operating element and the maximum force for seating and unseating the valve may be as per EN 12570, Clause 5.1 or the maximum force required to achieve a breakaway torque at the seat shall not exceed 360N.

Valve shall be provided with a gearbox if the allowable force and dimensional limits are exceeded.

The gearbox shall be of the self-locking gear type, dust-proof and weatherproof construction with output torque at least 1.5 times the maximum required operating torque of the valve.

The gearbox shall be provided with a position indicator

6.13. Actuators

If actuators are specified, valve shall be fitted with an actuator that is integral with the valve and supplied by the valve manufacturer according to the requirements specified in the datasheet.

Quarter-turn on/off actuators shall comply with DEP 32.36.01.18-Gen. and DEP 32.36.01.17-Gen Gear operated and actuated valves shall be fitted with a mounting flange in accordance with ISO 5211.

7. Quality Control

MANUFACTURER is required to submit with quotation his proposed inspection data sheet, which shall cover all requirements indicated in the Inspection Data Sheet and any other applicable referenced documents. Materials, design, manufacturing, and preparation for shipment of the Valves shall be subject to inspection by contractor and/or EMPLOYER or their appointed third-party inspection agency designated in the contract and, where applicable, by certification authority in the MANUFACTURER's manufacturing plant.

The MANUFACTURER shall assume overall responsibility for the design, engineering, fabrication, and testing procedures of the Valves and shall provide a complete guarantee to cover these activities. The MANUFACTURER shall maintain quality records of all tests, inspection and measurements detailed in this document and the Contractor approved quality plan as documentary evidence of conformance to quality requirements. Quality records shall be made available to Contractor's representative for analysis and review.

8. Quality Assurance and Quality Control

MANUFACTURER shall show that an effective quality assurance system is in operation for both product and services and this shall comply with ISO 9001 or equivalent as approved by EMPLOYER.

The MANUFACTURER shall establish and maintain a fully documented Quality Assurance /Quality Control system in accordance with ISO 9001 to ensure:

- Effective inspection and objective evidence that items conform to Purchase Order requirements
- Adequate identification and suitable handling of items.

The MANUFACTURER shall submit on request, documentary evidence of any Quality Reviews to ISO 9001, and internal system audits and corrective action requests.

All certification, test results, reports and any other documentation submitted to the EMPLOYER relative to the inspection system and results of inspections shall be in the English language.

Any verification of quality performed by EMPLOYER, or its representative will not relieve the MANUFACTURER in any way of its responsibility to produce acceptable items nor shall it preclude any subsequent rejection of items.

EMPLOYER shall establish mandatory hold points where applicable, which require verification of selected characteristics of any items or process by EMPLOYER and beyond which the manufacture shall not proceed until released.

The MANUFACTURER shall establish and maintain a system of traceability, which complies with ISO 9001. The system shall be such that each line-Valves is allocated a unique number, and this number shall be maintained such that at any point the individual line-Valves can be traced back to its original heat number. These numbers will be in blocks allocated by the EMPLOYER to the MANUFACTURER, and the MANUFACTURER shall demonstrate its ability to maintain traceability of such numbers throughout production.

9. Inspection and Testing

9.1. General

Prior to the commencement of manufacture, the MANUFACTURER shall prepare a written Inspection and Test Plan, which describes the inspection to be performed on each product. The Inspection and Test Plan, reference procedures and changes thereto are subject to the approval of the EMPLOYER. MANUFACTURER shall provide the following in addition to the Inspection and Test Plan:

i. A flowchart illustrating each manufacturing point affecting the quality of the product and its relative location in the production cycle where conformance of characteristics shall be verified. The MANUFACTURER shall include all inspection points of its own verification of quality.
ii. The characteristics to be inspected at each inspection point from procedural approval to shipping inspection. The procedures shall be submitted to EMPLOYER for review and approval prior to production commencement.

9.2. Quality Records

The MANUFACTURER shall maintain quality records of all tests, inspection and measurement's detailed in this document and the EMPLOYER approved quality plan as documentary evidence of conformance to quality requirements. Quality records shall be made available to EMPLOYER's representative for analysis and review.

Quality records shall include item identification by reference to revision number, acceptance criteria, specific inspection performed, and results obtained (If measurements are not a requirement, include in the record the basis of acceptance), date of inspection, identification of inspector, data recorder charts, qualification of material, personnel, procedures, and equipment.

The MANUFACTURER shall perform trend analyses of all factors affecting the quality of the product.

9.3. Final Inspection

The MANUFACTURER shall inspect the final item to ensure compliance with contract requirements. A check shall be made of all inspection records to verify that items were inspected at all points shown in the Inspection and Test Plan and that these records are complete and available to EMPLOYER's representative. Visual inspection and dimensional checks shall be performed both at the point of discharge and receipt to confirm that no damage took place during transportation.

9.4. Verification

All MANUFACTURER inspection systems may be subject to evaluation and surveillance by EMPLOYER's representative to ensure that they meet the requirements of this standard and

Purchase Order. The MANUFACTURER's operations required by this standard are subject to:

- Procedure compliance checking at unscheduled intervals to determine that the MANUFACTURER's inspection system is being effectively applied.
- Product verification to determine compliance with contract requirements. The method of verification is at the discretion of EMPLOYER's representative.

9.5. Accommodation and Assistance

The MANUFACTURER may be asked to provide EMPLOYER's representative with office accommodation and facilities required for the proper accomplishment of his work and shall provide any reasonable assistance required by EMPLOYER's representative for verification, documentation, or movement of items.

EMPLOYER's representative shall be afforded unrestricted opportunity to perform verification of inspection. The MANUFACTURER shall make available its inspection equipment for use by the EMPLOYER's representative to verify conformance to contract requirements.

The MANUFACTURER personnel shall be made available for the operation of inspection equipment as reasonably required by EMPLOYER.

9.6. NDE

All valves external and internal surfaces must be 100% visually inspected as per ASME V, Article 9. All non-destructive examination on casting is deemed performed on finished casting after all repair works and the complete heat treatment sequence. Acceptance criteria shall be as per MSS SP 55.

Requirements for weld repair shall be in accordance with ASME B16.34, Section 8.4.

9.7. Pressure Test

All valves (i.e., 100% of the supply) shall undergo pressure testing by the Manufacturer in his shop and under his responsibility in accordance with the requirements of API Spec 6D supplemented by DEP 31.36.00.30-Gen

The Manufacturer shall submit for the EMPLOYER's approval its detailed testing procedure for each type of valves, including each pressure class and size. This procedure must be in full accordance with the present specification requirements and the applicable codes.

The seating and sealing surfaces of valves design to permit sealant injection shall be free from the sealant before testing. The internals of valve shall be dry and clean.

Testing medium shall be clean, inhibited, fresh water. Valves with Stainless steel trims shall be tested using inhibited fresh water having a maximum chloride content of 30 ppm.

The temperature of the test must be a minimum of +10°C. In case of non-impact tested body materials, the minimum temperature of the test medium shall be +16°C.

After testing with water, all valves shall be thoroughly dried to prevent possible corrosion from the water.

9.8. Stem Backseat Test

Valve with stem backseat feature shall undergo a test at 1.5 times the pressure rating in accordance with the requirements, procedure, and acceptance criteria of API 6D para 10.2

9.9. Hydrostatic Shell Test

Hydrostatic shell test shall be performed on a fully assembled valve at 1.5 times the pressure rating in accordance with the requirements, procedure, and acceptance criteria of API 6D para 10.3

9.10. High Pressure Gas Seat

High pressure gas seat test shall be perform in lieu of hydrostatic seat test since valves are intended for gas service. An inert gas shall be used as the test medium.

For block valves with two seating surfaces, leakage from each seat shall be monitored through the body cavity connection. The seat test procedure based on the type of seating surfaces shall be in accordance with API 6D and DEP 31.36.00.30-Gen Annex H.4.3. Acceptable leakage for soft seated valves shall not exceed ISO 5208 rate A. (No visible leakage) while leakage rate for metal to metal seated valves shall not exceed ISO 5028 Rate D.

Valves with self-relieving seats shall be tested that it does not exceed 133 % of the pressure rating of the valve at its maximum specified design temperature.

9.11. Mechanical Test

The requirements of MESC SPE 77/302 for material toughness for pressure containing parts in valves, impact test for welding on valves and heat treatment shall be met for all valves

9.12. Test failure, re-work, re-test

In case a test fails, the Manufacturer is permitted to re-submit the same valve only once, and only after it has been properly re-worked and the reasons for failure thoroughly examined and explained by the Manufacturer to the Inspection Agency (who must report them). Should a given valve fail to pass a test for the second time, it shall be rejected and replaced by a new one.

9.13. Supplementary test

9.13.1. Torque/Thrust Testing

The maximum measured torque required to operate ball valve at specified pressure must be less than 75 % of the valve design torque.

The valve operating conditions for obtaining the torque values and other requirements in this test are specified in API 6D

10. Other Requirements

10.1. Painting

Surface preparation and application of painting and protective coatings shall be carried out in accordance with the project Painting and Coating for Above Ground Specification. Handle or steering shall be in the same material grade as the valve body to avoid galvanic corrosion.

The interior surfaces, the threaded parts, the bevelled or socket-welding ends shall be painted if not machined. The contact seal surfaces and valve tag identification plate shall not be painted.

10.2. Preservation and Shipping

Valves shall be properly preserved, fitted with end caps or flange protectors, placed on an adequate surface and packaged to ensure protection against damage, ingress of moisture, rainfall and dirt during outdoor storage for at least one year.

Machined surfaces shall be cleaned with a suitable solvent and immediately protected with an easily removable rust preventative coating or plastic film.

Valve preservation shall be guaranteed by the manufacturer for outdoor storage for at least one year.

Preparation for shipment shall be as per DEP 31.36.00.30-Gen

10.3. Marking

All valves shall be marked in accordance with table 12 of API 6D.

English language shall be used for all markings on nameplate and valve parts.

In addition, manufacturer shall follow the following requirements of DEP 31.36.00.30-Gen

- Body marking shall be made by stamping with a low-stress die stamp on a
- flange rim (for flanged valves) or on the valve body (for butt-weld valves).
- The size of the letters marked on name plate shall not be less than 3mm for valves NPS 2 and above Specification for nameplate and letter size for valves sizes less than NPS 2 shall be as per manufacturer’s standard.
- The nameplate(s) shall contain both metric and USC units.
- The seat design type SPE or DPE shall be marked on the flange end that contain the seat configuration.

10.4. Certification

Certification requirements shall comply with DEP 31.36.00.30-Gen. section 15.0

10.5. Documentation

10.5.1. General

The Vendor's documentation shall comply with the following:

- It shall be in the English language only.
- It shall be in SI units, although flange diameters shall be in nominal inch size.
- The minimum documentation to be provided by manufacturer are possibly prior to shipment;
- material test reports and inspection certificates, traceable by heat number to the foundry or mill;
- NDE reports, including sketches if necessary, showing the locations of examination traceable by heat or serial number.
- The retention period for NDE reports shall be 5 years.
- The retention period for radiographs shall be at least 12 months.)
- Listing of applicable and authorised concessions, waivers and/or material substitutions; m)
- listing of applicable manuals (e.g., assembly or maintenance manuals).
- Valve design and calculation notes.

10.5.2. Design drawings

All drawings shall be drawn or plotted onto A series paper sizes. Drawings may be either A3 or A4 sizes.

For each valve, or group of identical valves, the Vendor shall provide a General Arrangement (GA) drawing or drawing set. GA drawings shall show the outline of the valve, with appropriate sections to completely describe the arrangement. Drawings shall be fully dimensioned and provide detail on face- to-face and end preparations, tabulation showing material designation of all valve parts, design data and code and other details as per manufacturers documentation standards.